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Lecture 2 Statistical Machine Learning

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

1. Statistical Machine Learning
2. Assess Model Accuracy
3. Summary and Remark

Notations

- X: Input variable, predictors, independent variables, features, covariates, or sometimes just variables.
- Y: Output variable, response or dependent variable.

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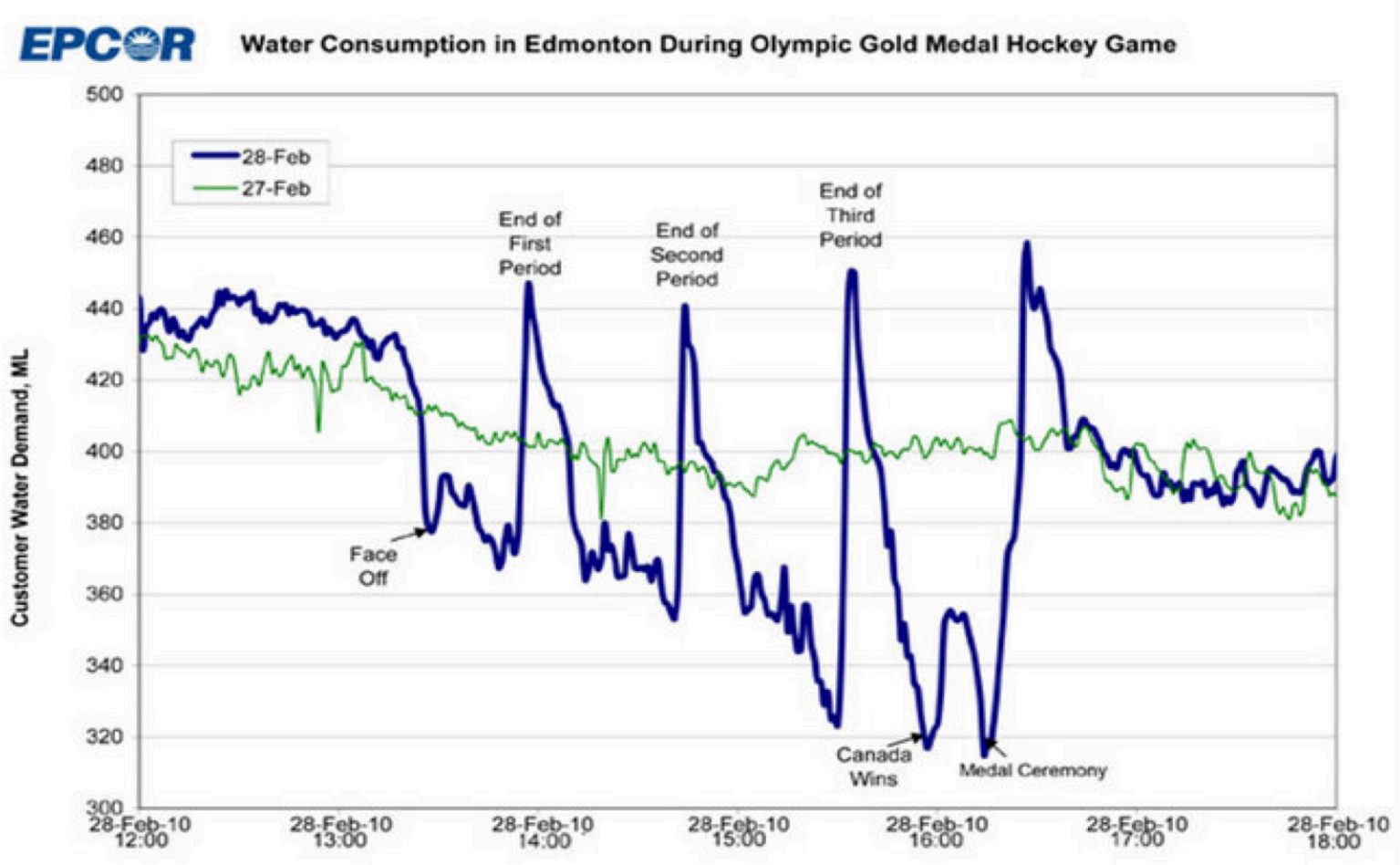
Seeing the data

- They say a picture is worth 1000 (10000) words



Vancouver 2010 final Canada vs. USA

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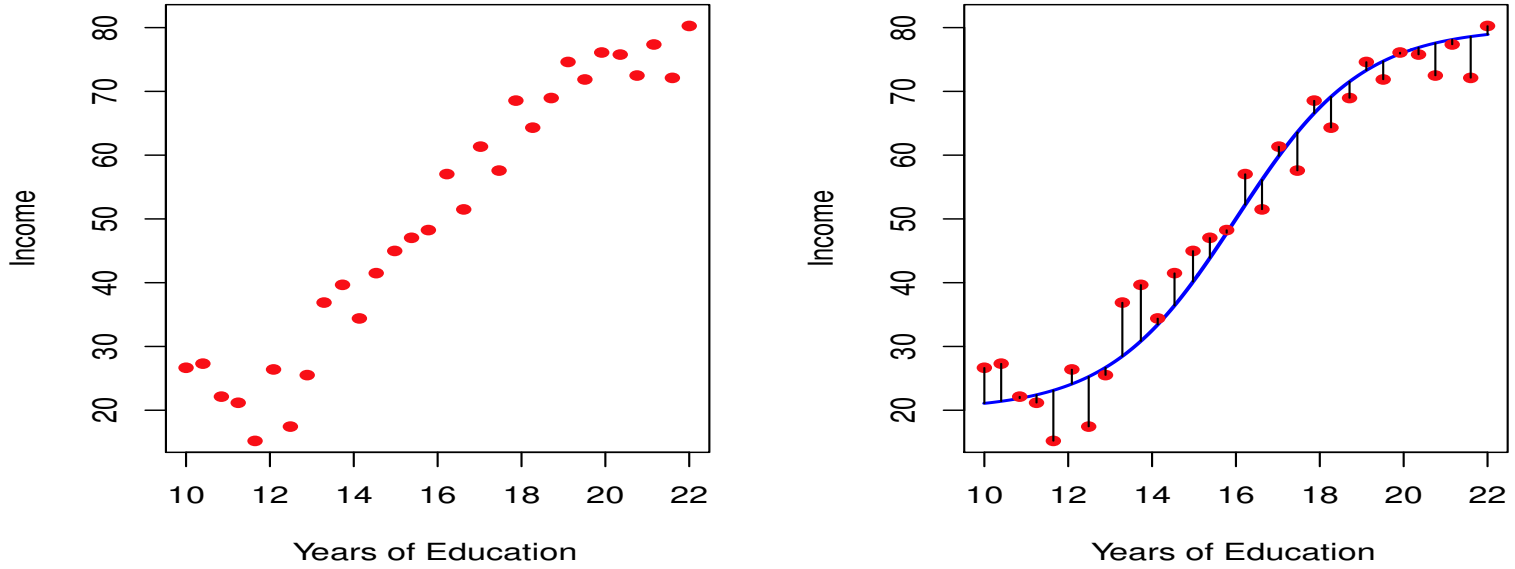
Statistical Machine Learning

- Given response Y_i and covariates $\boldsymbol{X}_i = (x_{1i}, x_{2i}, \cdots, x_{pi})^T$, we model the relationship

$$Y_i = f(\boldsymbol{X}_i) + \varepsilon_i,$$

where f is an unknown function and ε is random error with mean zero.

- A Simple example



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Estimate or learn the relationship

- Statistical machine learning is to estimate the relationship f, or using data to learn f. Why?
- To make prediction for the response Y for a new value of $\boldsymbol{X}$; → ML
- To make inference on the relationship between Y and $\boldsymbol{X}$, say, which x actually affect Y, positive or negative, linearly or more complicated. → Statistical learning
- Prediction Interested in predicting how much money an individual will donate based on observations from 90,000 people on which we have recorded over 400 different characteristics.
- Inference Wish to predict median house price based on 14 variables. Probably want to understand which factors have the biggest effect on the response and how big the effect is.

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Estimate or learn the relationship

- How estimate or learn f?
- Parametric methods say, linear regression (Chapter 3)

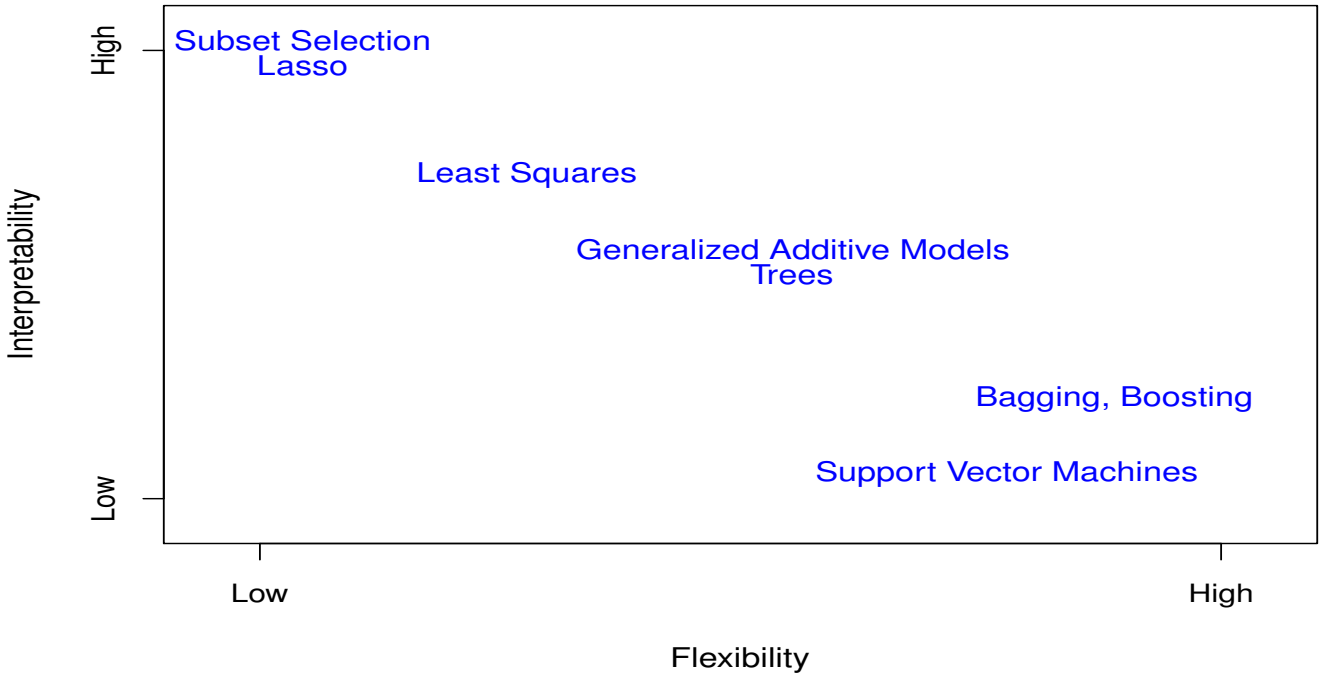
$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi},$$

by certain loss function, e.g. ordinary least squares (OLS).

- Nonparametric methods, say, spline expansion (Chapter 5) and kernel smoothing (Chapter 6) methods. (Now, we neural networks)
- Nonparametric methods are more felxible but need more data to obtain an accurate estimation.

Tradeoff between accuracy and interpretability

- The simpler, the better - parsimony or Occam's razor.
- A simple method is much easier to interpret, e.g. linear regression model.
- A simple model is possible to achieve more accurate prediction without overfitting. It seems counter intuitive though.



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Quality of fit (to get the performance)

Good -> Low MSE

- A common measure of accuracy is the mean squared error (MSE),

$$\text{MSE} = 1/n \sum_i \left(Y_i - \hat{Y}_i\right)^2,$$

Also can be $\frac{1}{n} \Sigma (\beta_i - \hat{\beta}_i)^2$

where $\hat{Y}_i$ is the prediction using the training data.

- In general, we minimize MSE and care how the method works for new data, we call it test data.
- More flexible models could have lower MSE for training data but higher test MSE.

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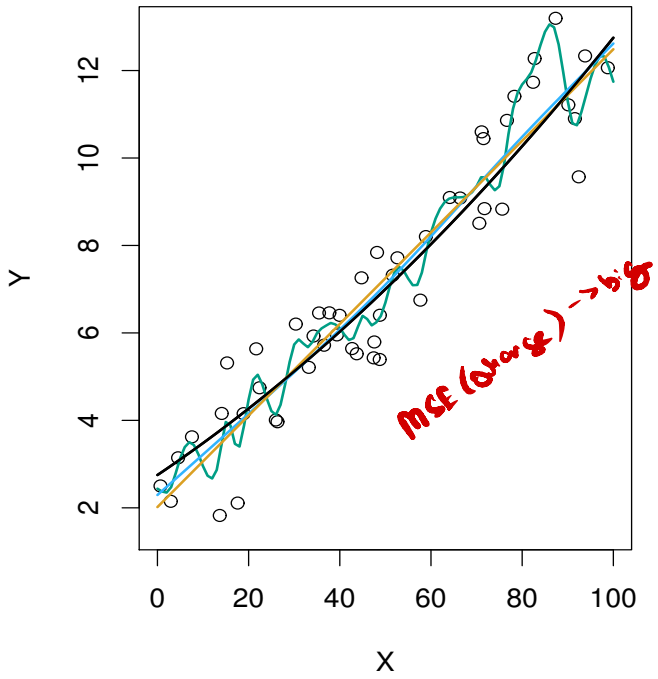
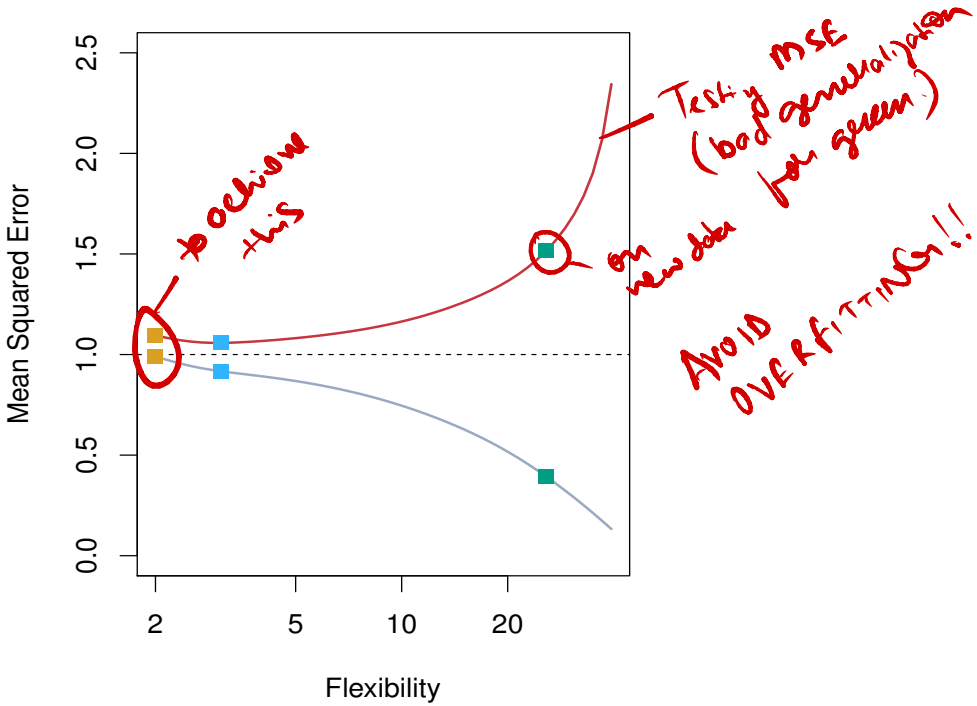
Levels of flexibility





- Black - Truth; Orange - Linear Estimate; Blue - smoothing spline; Green - smoothing spline (more flexible)
- RED - Test MSE; Grey - Training MSE; Dashed - Minimum possible test MSE (irreducible error)

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Bias and Variance tradeoff

- There are always two competing forces that govern the choice of learning method i.e. bias and variance.
- Bias refers to the error that is introduced by modeling a real life problem (that is usually extremely complicated) by a much simpler problem.
- The more flexible/complex a method is the less bias it will generally have.
- Variance refers to how much your estimate for f would change by if you had a different training data set.
- Generally, the more flexible a method is the more variance it has.

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Bias and Variance tradeoff

- For a new observation Y at $\boldsymbol{X} = \boldsymbol{X}_0$, the expected MSE is

$$E\left[\left(Y - \hat{Y}|\boldsymbol{X}_0\right)^2\right] = E\left[\left(f(\boldsymbol{X}_0) + \varepsilon - \hat{f}(\boldsymbol{X}_0)\right)^2\right] = \text{Bias}^2\left[\hat{f}(\boldsymbol{X}_0)\right] + \text{Var}\left[\hat{f}(\boldsymbol{X}_0)\right] + Var[\varepsilon].$$

- What this means is that as a method gets more complex the bias will decrease and the variance will increase but expected test MSE may go up or down!

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Bias-Variance trade-off

Consider the model: $Y = \beta_1 Z_1 + \cdots + \beta_r Z_r + \varepsilon$ (we skip the constant term for simplicity). Denote the estimate for β is $\hat{\beta}$.
We define MSE (mean of squared error):

$$\mathsf{MSE}(\hat{\beta}) = \mathbb{E}\left[\|\hat{\beta} - \beta\|^2\right],$$
$$\mathsf{Bias}(\hat{\beta}) = \mathbb{E}\hat{\beta} - \beta.$$

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Theorem

We have

$$\mathsf{MSE}(\hat{\beta}) = \left\|\mathsf{Bias}(\hat{\beta})\right\|^2 + \mathsf{Tr}(\mathrm{Cov}(\hat{\beta})).$$

$E[\hat{\beta}] = 0$?

Proof.

$$\begin{aligned}
\mathsf{MSE}(\hat{\beta}) =& \mathbb{E}\|\beta - \mathbb{E}\hat{\beta} + \mathbb{E}\hat{\beta} - \hat{\beta}\|^2 \\
=& \|\beta - \mathbb{E}\hat{\beta}\|^2 + 2\mathbb{E}\left\langle \beta - \mathbb{E}\hat{\beta}, \mathbb{E}\hat{\beta} - \hat{\beta}\right\rangle + \mathbb{E}\|\mathbb{E}\hat{\beta} - \hat{\beta}\|^2 \\
=& \left\|\mathsf{Bias}(\hat{\beta})\right\|^2 + 0 + \mathbb{E}\mathsf{Tr}\left((\mathbb{E}\hat{\beta} - \hat{\beta})'(\mathbb{E}\hat{\beta} - \hat{\beta})\right) \\
=& \left\|\mathsf{Bias}(\hat{\beta})\right\|^2 + \mathbb{E}\mathsf{Tr}\left((\mathbb{E}\hat{\beta} - \hat{\beta})(\mathbb{E}\hat{\beta} - \hat{\beta})'\right) \\
=& \left\|\mathsf{Bias}(\hat{\beta})\right\|^2 + \mathsf{Tr}\mathbb{E}\left((\mathbb{E}\hat{\beta} - \hat{\beta})(\mathbb{E}\hat{\beta} - \hat{\beta})'\right) \\
=& \left\|\mathsf{Bias}(\hat{\beta})\right\|^2 + \mathsf{Tr}(\mathrm{Cov}(\hat{\beta})).
\end{aligned}$$

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Summary and Remark

- What is Statistical Machine Learning
- How to assess the accuracy
- Read textbook Chapter 1 and 2